



E-COMMERCE PERFORMANCE ANALYSIS

Sales, Delivery Performance,
Customer Experience & Geographic Analysis



Power BI



Power Query



DAX



Data Modeling



Data Analysis



Using Olist Brazilian E-Commerce Public Dataset

A simulated business case for
portfolio purposes.

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1. Executive Summary

E-Commerce Performance Analysis

This case study evaluates the sales, delivery performance, customer experience, and geographic performance of an e-commerce business using the Olist Brazilian E-Commerce Public Dataset. The analysis was designed to identify key performance trends, understand factors associated with customer satisfaction, and highlight opportunities for operational and commercial improvement.

The analysis found that **91.87% of delivered orders were completed on time**, while late deliveries were associated with substantially lower customer satisfaction, with average review scores of **2.57 for late deliveries versus 4.29 for on-time deliveries**. Revenue was also geographically concentrated, with **São Paulo contributing approximately 38.3% of total revenue**, highlighting both the strength of the market and an opportunity to diversify revenue across other states.

Based on these findings, the analysis recommends **prioritizing delivery performance improvements**, particularly in weaker-performing states and categories with higher volumes of late deliveries, while maintaining São Paulo's strong position and exploring opportunities to grow higher-value customer segments across other markets.

Key Findings

- **91.87% on-time delivery rate** — delivery performance is strong overall, but approximately 8.13% of delivered orders were late.
- **1.72-point review score gap** — late deliveries had substantially lower average review scores than on-time deliveries.
- **38.3% of revenue from São Paulo** — revenue is concentrated in one major market, creating an opportunity for geographic diversification.
- **Category and state differences matter** — revenue, customer volume, revenue per customer, and delivery performance vary considerably across categories and states.
- **Commercial opportunities exist beyond overall revenue** — analyzing customer value and revenue per order helps identify markets and categories that may warrant targeted attention.

Recommendation

The primary recommendation is to improve delivery reliability while using geographic and category-level analysis to guide commercial priorities. Reducing late deliveries could improve the customer experience, while diversifying revenue beyond São Paulo and focusing on higher-value segments could support more balanced growth.

2. Business Scenario & Objectives

Business Scenario

For this portfolio project, I assumed the role of a **Data Analyst supporting an e-commerce business**. The business has transaction, customer, product, seller, payment, review, and delivery data but needs a clearer view of overall performance and the factors affecting customer experience.

Management wants to understand:

- How sales performance is changing over time
- Which product categories and markets contribute the most revenue
- How effectively orders are being delivered
- Whether delivery performance is associated with customer satisfaction
- Which geographic markets and product categories represent potential opportunities for improvement or growth

The analysis uses the **Olist Brazilian E-Commerce Public Dataset** and is presented as a simulated business case for portfolio purposes.

Business Objective

The primary objective was to transform the available transactional data into an interactive Power BI dashboard that helps management **understand performance, identify areas requiring attention, and support data-driven commercial and operational decisions**.

The analysis focuses on three major areas:

1. Commercial Performance

Evaluate revenue, order volume, customer value, product categories, and geographic markets.

2. Delivery & Customer Experience

Measure on-time delivery performance and examine the relationship between delivery outcomes and customer review scores.

3. Market Opportunities

Compare states and product categories based on market size, revenue contribution, customer value, and order economics.

Key Business Questions

The analysis was designed to answer the following questions:

1. **How is overall e-commerce performance changing over time?**
2. **Which product categories and states contribute the most revenue?**
3. **How large is the customer base across different markets?**
4. **Which markets generate higher revenue per customer?**
5. **What percentage of orders are delivered late?**
6. **Is delivery performance associated with customer satisfaction?**
7. **Which states have weaker delivery performance?**
8. **Which product categories account for the highest volume of late deliveries?**
9. **Which categories generate higher revenue per order?**
10. **Where should management prioritize operational or commercial attention?**

Scope

The analysis covers orders within the selected dashboard analysis period:

January 2017 – August 2018

The dashboard allows users to interactively adjust the analysis period using the shared date slicer.

3. Data & Preparation

3.1 Dataset

The analysis uses the **Olist Brazilian E-Commerce Public Dataset**, a publicly available e-commerce dataset containing information about orders, customers, products, sellers, payments, reviews, and delivery dates.

For this portfolio project, the dataset was treated as a simulated business data source. The objective was to transform the raw tables into a clean analytical model that could support sales, delivery, customer experience, and geographic analysis.

The preparation process was performed in **Power BI Power Query**, following the principle of cleaning and validating the data before building the analytical model.

3.2 Data Structure

The dataset contains multiple related tables representing different aspects of the e-commerce operation.

Core tables used

Table	Purpose
Orders	Order status, purchase dates, approval, delivery and estimated delivery dates
Order Items	Products purchased, sellers, prices, freight and shipping limits
Customers	Customer identifiers and geographic information
Products	Product identifiers and category information
Sellers	Seller identifiers and seller information
Order Reviews	Customer review scores and review information
Order Payments	Payment types and payment values

The tables were kept at their appropriate grain rather than combining them into one large table. This supported the later development of a structured star-schema model.

3.3 Initial Data Profiling

Before making transformations, the tables were profiled in Power Query to identify errors, missing values, duplicate values, and unusual distributions.

The profiling process included:

- Row and column counts
- Error values
- Empty/null values
- Distinct and unique values
- Minimum and maximum values
- Data types
- Numerical distributions where applicable

This follows the project approach of understanding the data and questioning inconsistencies before making changes, rather than blindly cleaning values.

Example: Customer IDs

The customer identifier field contained:

- **3,095 values**
- **0 errors**
- **0 empty values**
- **3,095 distinct values**
- **3,095 unique values**

This indicated that the identifier was suitable for use as a key within the relevant customer data.

Example: Product Categories

The product category field contained:

- **32,951 records**
- **623 empty values**
- **72 distinct categories**

The missing category values were investigated rather than automatically replaced.

The related category profile showed that **610 of the 623 missing entries were empty strings**, while only a small number contained actual category values. This indicated that the missing-category issue was primarily a data-completeness problem rather than a large number of inconsistent category labels.

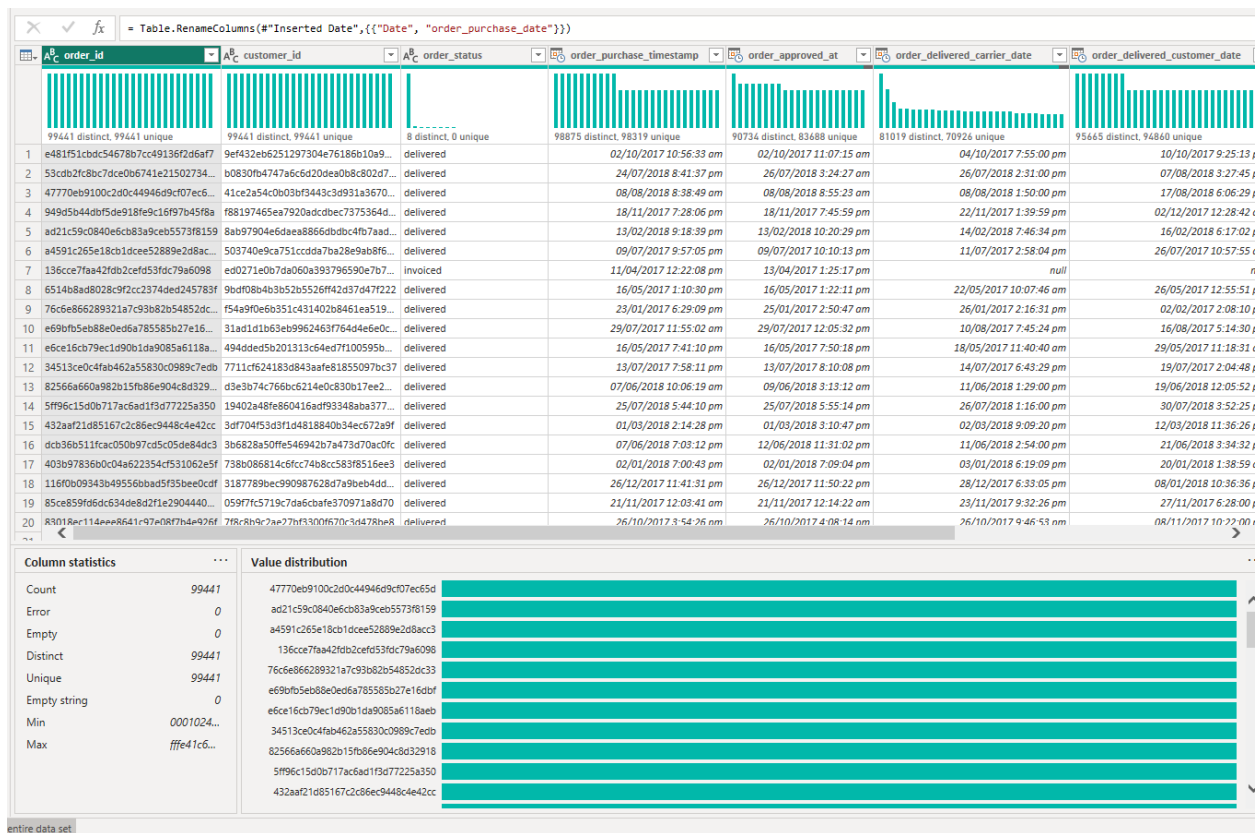


Figure 1 - Initial Data Profiling in Power Query

3.4 Data Type Corrections

One of the preparation steps was reviewing and correcting the data types of the imported columns.

Examples included:

Orders

Column	Data Type
order_id	Text
customer_id	Text
order_status	Text

Column	Data Type
order_purchase_timestamp	Date/Time
order_approved_at	Date/Time
order_delivered_carrier_date	Date/Time
order_delivered_customer_date	Date/Time
order_estimated_delivery_date	Date/Time

Order Items

Column	Data Type
order_id	Text
order_item_id	Whole Number
product_id	Text
seller_id	Text
shipping_limit_date	Date/Time
price	Decimal Number
freight_value	Decimal Number

Correct data types were important because the analysis relies heavily on **date comparisons, numeric aggregation, identifiers, and relationships between tables.**

3.5 Data Quality Considerations

Several data-quality observations were identified during profiling.

Missing product categories

Some products did not have a populated category value. Rather than inventing categories, the missing values were retained/handled explicitly so that the analysis would not introduce unsupported assumptions.

Date completeness

Delivery-related analysis required distinguishing between orders that had a customer delivery date and those that did not.

This became particularly important when calculating:

On-Time Delivery %

Only orders with a valid customer delivery date were considered delivered for the delivery-performance analysis.

Identifier integrity

Key identifier columns were checked for errors, empty values, and uniqueness where appropriate before being used in relationships.

3.6 Preparation Outcome

After profiling and transformation, the data was prepared for modeling with:

- Correct data types
- Identifiable keys
- Investigated missing values
- Consistent date fields
- Clean numerical fields
- Appropriate table structure

The cleaned tables were then used to build the Power BI data model.

The overall preparation approach followed the project's core principle:

Clean before modeling.

This was important because the quality of the subsequent DAX calculations and dashboard insights depends on the underlying data and model being reliable.

4. Data Model

The cleaned data was structured in Power BI using a **star-schema approach**, separating transactional fact tables from descriptive dimension tables. This structure was designed to make the model easier to understand, maintain, and use for DAX calculations.

4.1 Fact Tables

The main transactional tables were retained at their appropriate grain.

Orders Fact

Contains one record per order and provides the foundation for order-level analysis, including:

- Order status
- Purchase date
- Approval date
- Carrier delivery date
- Customer delivery date
- Estimated delivery date
- Customer identifier

Order Items Fact

Contains individual items associated with orders and supports:

- Revenue analysis
- Product/category analysis
- Seller analysis
- Freight analysis
- Order-level item analysis

Other transactional tables, including **Order Payments** and **Order Reviews**, provide additional measures for payment and customer-experience analysis.

4.2 Dimension Tables

Descriptive attributes were separated into dimensions used to filter and analyze the transactional data.

Key dimensions include:

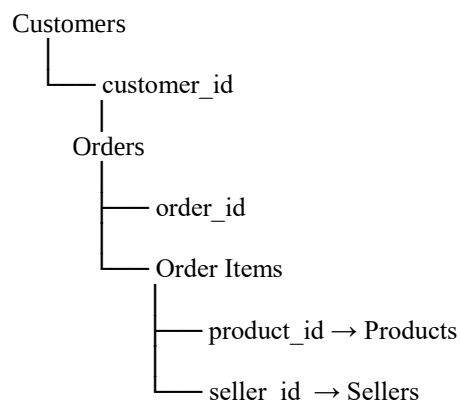
- **Customers** — customer identifiers and geographic attributes such as state
- **Products** — product identifiers and product categories
- **Sellers** — seller identifiers and seller attributes
- **Date** — calendar attributes used for time-based analysis

This separation allows the same dimensions to be used consistently across different analytical views.

4.3 Relationships

Relationships were created using the appropriate identifiers between the tables.

Examples include:



The model was designed so that descriptive dimensions could filter the relevant transactional data without unnecessarily combining tables into a single flattened dataset.

4.4 Why the Model Matters

The data model was an important part of the analysis because several business metrics required information from different tables.

For example:

Revenue by Product Category

requires product-category information from the Products table to filter transactional revenue from Order Items.

Similarly:

Delivery Performance

requires delivery dates from Orders, while:

Customer Experience

uses review scores from Order Reviews.

A structured model allows these different analytical areas to work together while preserving the original table grains.

4.5 Modeling Considerations

During development, filter context was an important consideration when creating measures.

For example, category-level analysis required ensuring that a product-category selection correctly affected the relevant order-level calculations. This led to additional measures for category-specific analysis rather than changing the model's relationships simply to force filtering behavior.

This approach helped preserve the integrity of the underlying model while allowing the dashboard to answer more specific business questions.

4.6 Model Outcome

The final model provided a foundation for:

- Revenue and order analysis
- Customer analysis
- Product/category analysis
- Seller analysis
- Delivery performance
- Customer review analysis
- Geographic analysis
- Time-based trends

The resulting structure allowed the dashboard to remain interactive while keeping the analytical logic separated from the presentation layer.

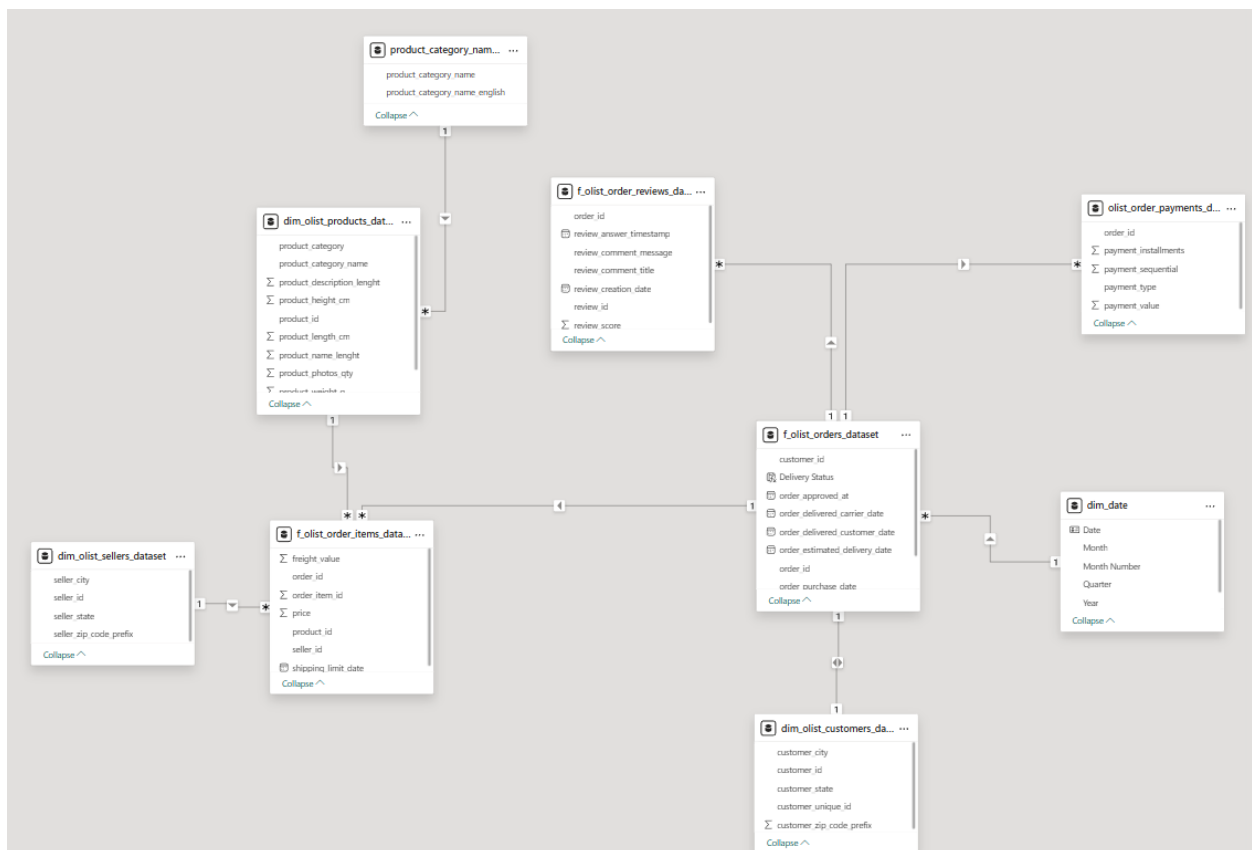


Figure 2 — Power BI Data Model

5. Analytical Approach & DAX

After preparing the data and establishing the model, the next step was to define the business metrics used throughout the analysis.

The analytical approach focused on creating **reusable DAX measures** rather than relying heavily on calculated columns. This allowed the metrics to respond dynamically to the dashboard's date, geographic, product, and delivery filters.

This follows the project's core DAX approach of focusing on **measures, filter context, and CALCULATE behavior**, while validating that each calculation produces a meaningful business result.

5.1 Core Business Measures

The following measures were developed to support the dashboard.

Total Revenue

Total Revenue =
SUM(f_olist_order_items_dataset[price])

This measure represents the total product revenue within the current filter context.

Total Orders

Total Orders =
DISTINCTCOUNT(f_olist_orders_dataset[order_id])

Orders are counted using distinct order_id values to avoid counting individual order items as separate orders.

Total Unique Customers

Total Unique Customers =
DISTINCTCOUNT(f_olist_orders_dataset[customer_id])

This measures the number of distinct customers represented within the current analysis context.

Revenue per Order

Revenue per Order =
DIVIDE(
 [Total Revenue],
 [Total Orders]
)

This provides an overall view of the average revenue generated per order.

5.2 Delivery Performance Measures

Delivery performance was analyzed by comparing the actual customer delivery date with the estimated delivery date.

A delivery was classified as **On-Time** when:

Actual delivery date $\leq$ Estimated delivery date

Orders without a customer delivery date were excluded from the delivered-order calculation.

On-Time Deliveries

```
On-Time Deliveries =  
CALCULATE(  
    COUNTROWS(f_olist_orders_dataset),  
    f_olist_orders_dataset[Delivery Status] = "On-Time"  
)
```

Delivered Orders

```
Delivered Orders =  
CALCULATE(  
    COUNTROWS(f_olist_orders_dataset),  
    NOT ISBLANK(  
        f_olist_orders_dataset[order_delivered_customer_date]  
    )  
)
```

On-Time Delivery %

```
On-Time Delivery % =  
DIVIDE(  
    [On-Time Deliveries],  
    [Delivered Orders]  
)
```

This resulted in an overall on-time delivery rate of approximately **91.87%** for the selected analysis period.

5.3 Customer Experience Measures

Customer review scores were used as a proxy for customer satisfaction.

Average Review Score

```
Average Review Score =  
AVERAGE(  
    folist_order_reviews_dataset[review_score]  
)
```

The analysis also compared review scores between on-time and late deliveries.

This produced:

Delivery Status Average Review Score

On-Time **4.29 / 5**

Late **2.57 / 5**

The difference was approximately:

1.72 points

This relationship was treated as an **association rather than proof of causation**.

5.4 Revenue per Customer

Revenue per customer was calculated as:

```
Revenue per Customer =  
DIVIDE(  
    [Total Revenue],  
    [Total Unique Customers]  
)
```

This measure was particularly useful for the geographic analysis because it allowed markets to be evaluated based on **customer value**, rather than revenue alone.

This helped distinguish between:

- Large markets with many customers
- Smaller markets with higher customer value
- Markets with both high scale and high customer value

5.5 Geographic Analysis

Three complementary metrics were used to evaluate state-level performance:

Market value

Revenue by State

Market size

Unique Customers by State

Customer value

Revenue per Customer by State

Using all three measures prevented the analysis from relying solely on total revenue.

For example, a state with high revenue may simply have a large customer base, while another state may generate less total revenue but have substantially higher revenue per customer.

This comparison helped identify potential opportunities for geographic diversification.

5.6 Category-Level Analysis

Product categories were analyzed using revenue, late-delivery volume, and revenue per order.

A category-specific revenue-per-order measure was required because the product category filter operates at the order-item level:

```
Category AOV =  
DIVIDE(  
  [Total Revenue],  
  DISTINCTCOUNT(  
    f_olist_order_items_dataset[order_id]  
  )  
)
```

This measure represents the revenue generated per order containing the selected category.

The distinction was important because an order can contain multiple items, and therefore **overall business AOV and category-level revenue per order are not necessarily equivalent metrics.**

5.7 Filter Context & Validation

One of the more important analytical considerations during development was understanding **filter context**.

Some dimensions, such as product category, operate through the Order Items table, while other measures depend on the Orders table. This required careful validation to ensure that category and geographic selections produced meaningful results.

Rather than changing relationships unnecessarily, additional measures were created where specific analytical contexts required different calculations.

This reflects the project's core principle:

Always ask: "What filter context am I controlling?"

Each major measure was validated against known totals and dashboard results before being used in the final visuals.

5.8 Analytical Outcome

The final DAX layer provided the measures needed to answer the original business questions across three analytical areas:

Commercial

- Revenue
- Orders
- Revenue per Order
- Revenue per Customer

Operational

- Delivered Orders
- On-Time Deliveries
- On-Time Delivery %
- Late Delivery %

Customer Experience

- Average Review Score
- Review Score by Delivery Status

- Review Score Gap

These measures formed the analytical foundation for the three-page Power BI dashboard.

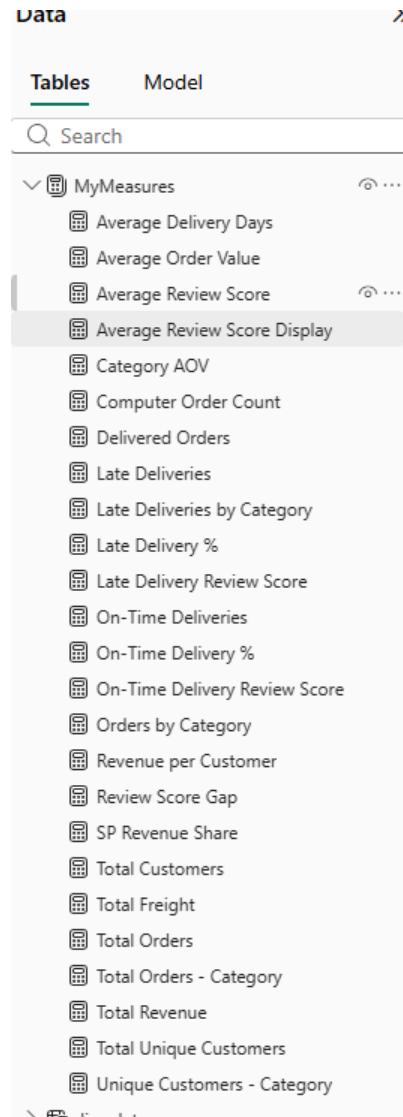


Figure 3— Core DAX Measures Used in the Analysis

6.1 Executive Overview

The first dashboard page provides a high-level view of overall business performance across revenue, orders, delivery performance, and customer experience.

Key Performance Indicators

For the January 2017–August 2018 analysis period:

Metric	Result
Total Revenue	₱13.54M
Total Orders	99.1K
Revenue per Order	₱136.66
On-Time Delivery	91.87%
Average Review Score	4.11 / 5

Revenue Trend

Revenue increased substantially throughout 2017 before becoming more volatile during 2018. The monthly trend indicates continued commercial activity, while also suggesting that performance should be evaluated alongside operational and customer-experience measures rather than revenue alone.

Category Performance

Revenue was concentrated among several major product categories, with **Health & Beauty** representing the highest-revenue category in the dashboard.

This indicates that category-level performance can provide useful direction for inventory, marketing, and commercial planning.

Key Business Finding

The executive overview indicates that the business has a strong overall commercial base, but **delivery performance and geographic concentration represent important areas for further investigation.**

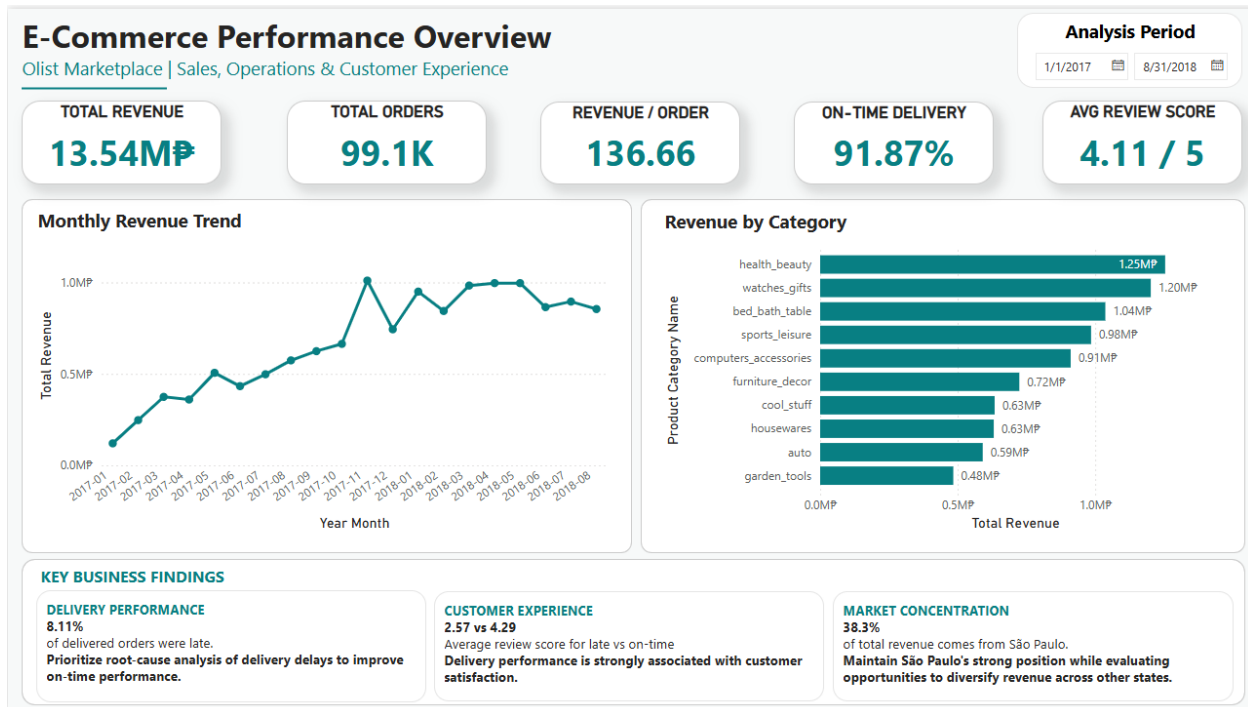


Figure 4— Executive Overview Dashboard

6.2 Delivery & Customer Experience

The second dashboard page focuses on the relationship between operational delivery performance and customer experience.

On-Time Delivery

Approximately:

91.87% of delivered orders were completed on time.

This means approximately:

8.13% were delivered late.

While the majority of orders were delivered on time, the late-delivery segment represents a meaningful operational issue because of its relationship with customer reviews.

Review Score Difference

The most significant customer-experience finding was the difference between review scores for on-time and late deliveries.

Delivery Status	Average Review Score
On-Time	4.29 / 5
Late	2.57 / 5

This represents a difference of approximately:

1.72 points

Late deliveries were therefore **associated with substantially lower customer review scores**.

The analysis deliberately describes this as an association rather than claiming that late delivery alone caused the lower ratings.

Review Distribution

The review-score distribution provides additional context to the average-score comparison. The distribution shows that the difference is not simply a small change in the average; the composition of customer ratings also differs between on-time and late deliveries.

This strengthens the case for treating delivery reliability as an important customer-experience metric.

Geographic Delivery Performance

The state-level analysis shows that delivery performance varies across markets.

Rather than treating delivery performance as a single company-wide metric, the dashboard allows management to identify **lower-performing states that may warrant operational investigation**.

Category Concentration of Late Deliveries

The analysis also identifies product categories generating a larger volume of late deliveries.

These categories should not automatically be interpreted as the *cause* of delays. Instead, they provide areas for further investigation into potential differences in seller behavior, fulfillment processes, product characteristics, or logistics.

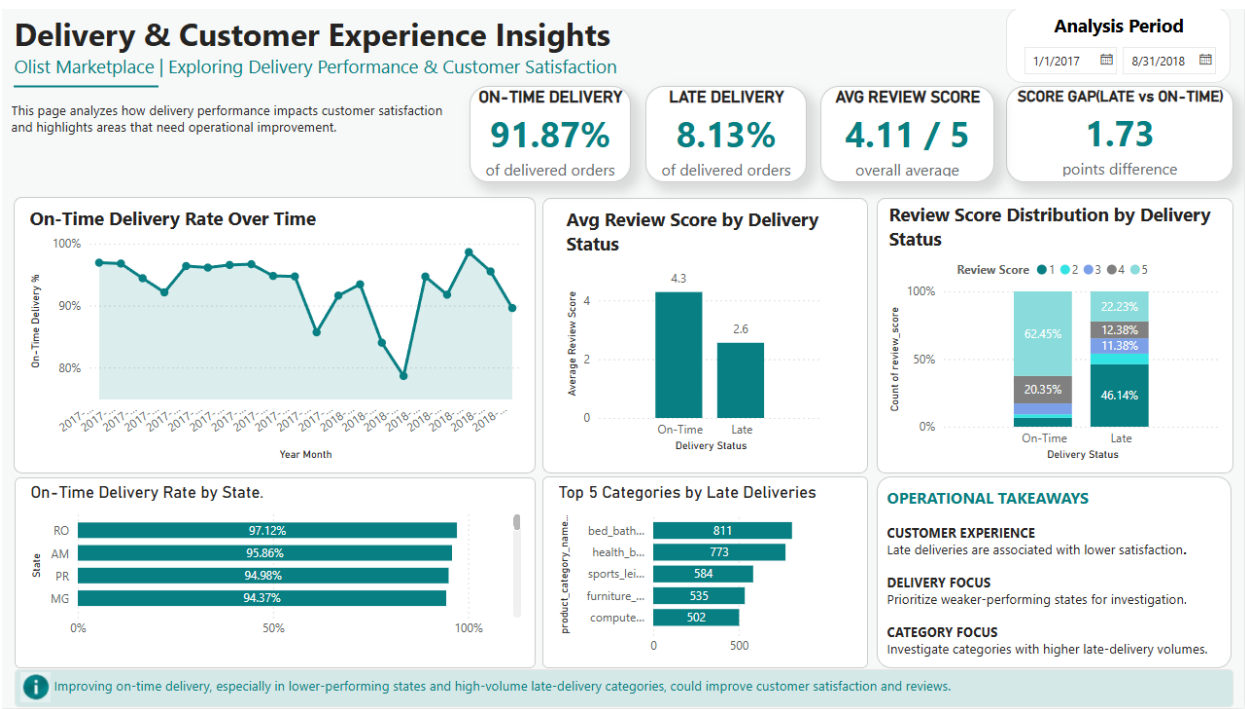


Figure 5— Delivery & Customer Experience Dashboard

6.3 Commercial & Geographic Performance

The third dashboard page examines where revenue comes from, how large different markets are, and how valuable customers are across states and product categories.

Geographic Revenue Concentration

One of the strongest commercial findings was:

São Paulo contributes approximately 38.3% of total revenue.

This demonstrates the importance of São Paulo as a core market.

However, the concentration also creates an opportunity to evaluate whether revenue growth can be diversified across other states.

Market Size vs. Market Value

The geographic analysis compares three dimensions:

Market value

Revenue by State

Market size

Unique Customers by State

Customer value

Revenue per Customer by State

This prevents the analysis from relying on revenue alone.

A state with high revenue may simply have a larger customer base, while another market may have fewer customers but generate substantially more revenue per customer.

Revenue vs. Order Volume

The scatter analysis compares revenue and order volume across the top states.

This helps identify different market profiles:

- High order volume + high revenue → **core markets**
- High order volume + relatively lower revenue → **potential value opportunity**
- Lower order volume + relatively high revenue → **higher-value niche markets**
- Low volume + low revenue → **lower immediate priority**

This provides a more nuanced view of geographic performance than simply ranking states by revenue.

Category Revenue per Order

The category analysis also showed significant variation in revenue generated per order across product categories.

Categories such as **Computers, Small Appliances, and Home Appliances** showed relatively high revenue per order within the selected analysis context.

These categories may provide opportunities for targeted commercial strategies, although the analysis does not by itself establish why these categories generate higher order values.

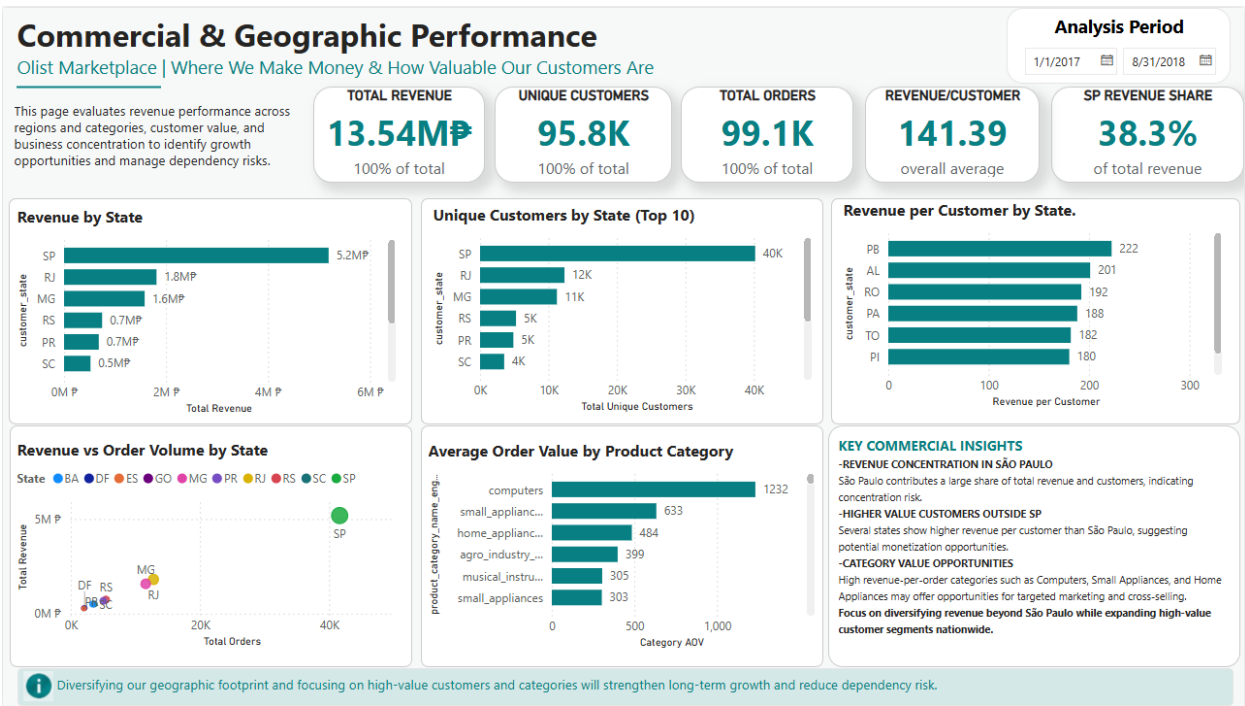


Figure 6— Commercial & Geographic Performance Dashboard

7. Recommendations

Based on the analysis across sales, delivery performance, customer experience, and geographic performance, the following recommendations are proposed.

7.1 Prioritize Delivery Reliability

The strongest finding from the analysis is the relationship between delivery performance and customer satisfaction.

Late deliveries represented approximately **8.13% of delivered orders**, while average review scores were substantially lower for late deliveries (**2.57**) compared with on-time deliveries (**4.29**).

Recommendation

Prioritize investigation of the operational factors associated with late deliveries, particularly in:

- Lower-performing states
- Product categories with higher volumes of late deliveries
- Areas where delivery performance is consistently below the overall benchmark

The objective should be to identify whether delays are related to fulfillment processes, seller performance, logistics, or other operational factors.

Expected business benefit

Improving delivery reliability could support:

- Higher customer satisfaction
- Fewer negative reviews
- Improved customer retention
- More consistent service quality

The analysis demonstrates an association between delivery performance and review scores, but further operational data would be required to establish causality.

7.2 Diversify Geographic Revenue

São Paulo represents approximately **38.3% of total revenue**, making it a critical market for the business.

Recommendation

Maintain São Paulo as a priority market while identifying opportunities to increase revenue contribution from other states.

This could include:

- Targeted customer acquisition
- Regional marketing campaigns
- Retention initiatives
- Category-specific promotions
- Expansion of successful product categories into promising markets

The objective is **not to reduce focus on São Paulo**, but to reduce excessive dependence on a single geographic market.

Expected business benefit

A more diversified revenue base could:

- Reduce geographic concentration risk
- Create additional growth opportunities
- Improve resilience across regional markets

7.3 Target High-Value Customer and Market Segments

The geographic analysis showed that revenue, customer volume, and revenue per customer vary across states.

Recommendation

Prioritize states that demonstrate attractive combinations of:

- Customer base size
- Revenue contribution
- Revenue per customer

- Delivery performance

Rather than evaluating markets solely by total revenue, management should consider **market size and customer value together**.

Expected business benefit

This approach can help identify markets where additional investment may generate stronger commercial returns.

7.4 Focus on High-Value Product Categories

The category analysis showed differences in revenue generated per order, with categories such as **Computers, Small Appliances, and Home Appliances** demonstrating relatively high revenue per order in the analyzed data.

Recommendation

Evaluate high-value categories for targeted commercial strategies such as:

- Cross-selling
- Bundling
- Targeted promotions
- Customer segmentation
- Regional expansion

However, category-level revenue per order should be considered alongside order volume and delivery performance before making inventory or marketing decisions.

7.5 Use a Balanced Decision Framework

The analysis suggests that management should avoid optimizing for a single KPI.

A stronger decision framework would evaluate:

Revenue + Customer Value + Market Size + Delivery Performance + Customer Experience

For example, a state with high revenue but poor delivery performance may require a different strategy from a smaller state with strong customer value and reliable delivery.

This approach allows commercial growth and operational improvement to be considered together.

7.6 Recommended Priorities

Based on the analysis, I would prioritize the actions as follows:

Priority	Action	Rationale
1	Improve delivery reliability	Strong association with customer satisfaction
2	Investigate weaker-performing states/categories	Identify operational improvement opportunities
3	Diversify revenue beyond São Paulo	Reduce geographic concentration
4	Develop high-value markets/categories	Support targeted commercial growth

8. Limitations & Next Steps

8.1 Limitations

Several limitations should be considered when interpreting the analysis.

Simulated Business Context

This project uses the Olist Brazilian E-Commerce Public Dataset as a **portfolio case study**. The business scenario, stakeholder requirements, and recommendations are simulated rather than based on direct access to an organization's internal objectives or operational processes.

Dataset Scope

The analysis is limited to the fields and time period available in the dataset. Additional operational information—such as carrier performance, warehouse processing times, seller fulfillment behavior, shipping routes, and logistics costs—was not available for this analysis.

Delivery Root Causes

The dataset allows delivery performance to be classified as **on-time or late**, but it does not provide a definitive field identifying the reason for each delay.

Therefore, categories or states with higher late-delivery volumes should be treated as **areas for investigation**, rather than proof of the underlying cause.

Correlation vs. Causation

The analysis identified a substantial difference in review scores between late and on-time deliveries.

However, the analysis does not establish that late delivery **caused** lower review scores.

Other factors—including product issues, seller performance, customer expectations, or order characteristics—may also influence customer satisfaction.

Geographic Concentration

São Paulo's approximately **38.3% revenue contribution** indicates significant revenue concentration, but concentration alone does not mean the market is underperforming.

São Paulo may simply represent a naturally larger or more mature customer market. Further analysis would be required before recommending a change in investment levels.

8.2 Recommended Next Steps

The analysis could be extended with additional data and deeper investigation.

1. Investigate Delivery Root Causes

Combine delivery data with:

- Seller performance
- Carrier information
- Shipping distance
- Processing time
- Warehouse/fulfillment information

This would help identify the actual drivers of late deliveries.

2. Analyze Customer Retention

Add customer-level behavioral analysis such as:

- Repeat purchase rate
- Customer lifetime value
- Time between purchases
- First-time vs repeat customers

This would determine whether delivery problems affect longer-term customer behavior.

3. Evaluate Profitability

Revenue does not equal profit.

Adding:

- Product cost
- Shipping cost
- Marketing cost
- Discounts
- Seller fees

would allow the analysis to move from **revenue performance** toward **true profitability**.

4. Develop Predictive Analysis

Historical delivery and order characteristics could potentially be used to identify orders at higher risk of late delivery.

This could allow the business to move from:

"Which orders were late?"

toward:

"Which orders are likely to be late, and what can we do before the delay occurs?"

9. Conclusion

This project demonstrates how Power BI can be used to transform raw e-commerce data into a structured decision-support tool.

The analysis found that the business had a strong overall commercial base, with **₹13.54M in revenue and 99.1K orders** during the selected analysis period. However, the analysis also identified meaningful opportunities around delivery reliability, customer experience, and geographic concentration.

The most significant finding was the substantial difference in customer review scores between on-time and late deliveries. Combined with the concentration of approximately **38.3% of revenue in São Paulo**, this suggests that the business should balance **operational improvement with geographic and commercial diversification**.

The resulting Power BI dashboard provides management with an interactive way to explore these patterns by time, state, delivery status, and product category.

More importantly, the project demonstrates the complete analytical workflow:

Business Question → Data Preparation → Data Modeling → DAX → Analysis → Visualization → Recommendation

This end-to-end approach was the primary objective of the portfolio project.